

ShegerShop Mobile Application Roadmap

Full marketplace app for Ethiopia — buyers, sellers, and couriers

Approach: Capacitor + Next.js hybrid **Horizon:** 2–4 months to market **Status:** Foundation complete (Sep 2026)

Doc date: September 6, 2026

Recommendation: Wrap the existing Next.js app with Capacitor (Android first, then iOS). Reuse ~80% of the codebase. Do *not* rewrite in React Native for v1.

1. Strategy comparison

Approach	Time to market	Code reuse	Fit for Ethiopia
Capacitor + mobile-ready API (recommended)	2–4 months	High (~80%)	Strong — Telebirr deep links, phone auth, seller/courier scan
PWA only	4–6 weeks	Highest	Good for validation; weak store presence
Full React Native rewrite	6–12+ months	Low	Only with a dedicated native team

2. Foundation deliverables (Phase 1) — completed

Item	Status	What shipped
Mobile JWT auth + Bearer tokens	Done	POST /api/auth/mobile/login & refresh; cookie or Bearer on protected routes
Buyer catalog API	Done	GET /api/products, GET /api/products/[id] (paginated, absolute image URLs)
Server-side cart	Done	Prisma Cart/CartItem; /api/cart; guest merge on login
Cloud uploads (S3/R2)	Done	Object storage with local-disk fallback; HTTPS URLs when configured
Payment deep links	Done	shegershop://payment/result; Telebirr/Chapa InAppBrowser; COD unchanged
Capacitor Android/iOS shell	Done	Remote server URL mode; Preferences token storage; deep-link intent filters
Ethiopia polish baseline	Done	Amharic UI path, FCM stub, 20-item pagination, Play Store beta guide

3. Three-phase roadmap

Phase 1 — Mobile backend foundation (4–6 weeks)

Make the REST layer work for a native HTTP client, not only browser cookies.

- Token auth (access + refresh JWT)
- JSON product catalog for buyers
- Server cart with cross-device sync
- S3-compatible product/license uploads
- Telebirr/Chapa return via app URL scheme

Phase 2 — Capacitor app shell (3–4 weeks)

- PWA manifest, icons, safe-area CSS
- Capacitor Android + iOS loading production HTTPS URL (fast updates without store wait)
- Role-based home after login: **BUYER** → shop/cart/orders · **SELLER** → /seller + scan · **DELIVERY** → /delivery + scan · **ADMIN** → scan only in app
- **Google Play first** (Android dominates Ethiopia); iOS second

Phase 3 — Ethiopia market optimizations (ongoing)

3G performance

- WebP thumbnails in API responses
- Default page size 20
- Optional offline last-viewed catalog cache

Localization & trust

- Amharic (Afaan Oromo optional)
- 09xx / +251 phone UX
- Prominent COD; Telebirr default online; Chapa backup

Notifications

- SMS already wired for order updates
- FCM push for in-app alerts
- SMS fallback when data is off

Tracking (future)

- QR on print labels
- Public /track/ [code] page
- Scan flow already built for sellers/couriers

4. Suggested timeline

Month	Deliverable
1	Mobile token auth + product APIs + server cart
2	Cloud uploads + payment deep links + Capacitor Android beta
3	Play Store beta (Addis); seller/courier scan QA on real devices
4	Amharic polish, push notifications, iOS, public launch

5. Immediate next steps (ops)

1. Deploy Next.js to HTTPS with Telebirr/Chapa credentials and run `db:migrate:deploy`
2. Set `CAPACITOR_SERVER_URL` to production; run `bun run cap:sync`
3. Open Android Studio (`bun run cap:open:android`), build signed AAB
4. Upload to Google Play Internal/Closed testing (Addis cohort)
5. Device QA: login (buyer/seller/courier), catalog on 3G, Telebirr round-trip, COD, package scan

6. Success metrics (Ethiopia)

- Checkout completion on mobile (Telebirr + COD)
- Seller time from order → labeled package (`mark_ready` scan)
- Courier claim-to-deliver time via scan
- First catalog load: < 3s on 4G, < 8s on 3G (target)

7. What not to do first

- Full React Native rewrite — duplicates seller/courier/scan work already shipped
- Full admin portal in v1 mobile — keep admin on web; scan tool in app is enough
- Offline-first everything — start online-first + light cache